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UPI FRAUD DETECTION USING MACHINE LEARNING

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Abstract—UPI has dramatically increased as upi becomes the most popular mode of payment for both online as well as regular purchase, cases of fraud associated with it are also rising. In this model sequencial operations in upi transaction processing using a Graphic Neural Network and XGBoost and Isolation Forest to show how it can be used for the detection of frauds. An gnn is initially trained with the normal behavior of a upi user. If an incoming upi transaction is not accepted by the trained gnn with sufficiently high probability,it is considered to be fraudulent.At same time, project ensure that genuine transactions are not rejected.Detailed effective results of this project compared with other techniques available in the literature.

Index Terms-UPI trasactions,gnn-graphical nueral networks,XGBooster classifier,fraud,Random forest,isolation forest,frame work.

I. Introduction

The Unified Payments Interface (UPI), which offers a quick and easy way to transfer money, has completely altered India's digital payment environment. However, due to its extensive use, there has been an increase in fraudulent activity, endangering user security and confidence. The complexity of fraudulent schemes and the volume of transactions involved in upi transactions may make it difficult to detect fraud. This study aims to minimize fraudulent transactions by identifying fraudulent transactions using powerful machine learning algorithms. Machine learning algorithms can accurately discriminate between legitimate and fraudulent operations by analyzing transaction trends and user behavior. In order to identify the best strategy, this study looks at and evaluates a variety of methods, including decision trees, random forests, logistic regression, and neural networks. The goal of this project is to use machine learning techniques to identify fraudulent transactions within the upi.Traditional rule-based systems often fall short in detecting complex and evolving fraud patterns. ML algorithms, on the other hand, can learn from vast amounts of transaction data, identifying subtle patterns and anomalies that indicate fraudulent behavior. These algorithms can be trained on historical data to recognize signs of fraud, such as unusual transaction amounts, locations, or frequencies.By analyzing these patterns, ML models can predict the likelihood of a transaction being fraudulent and flag it for review or block it altogether. The benefits of using ML for upi fraud detection are numerous. For one, it improves security by enhancing the detection and prevention of upi fraud. This, in turn, reduces financial losses for users and banks, which can occur due to unauthorized transactions. Moreover, a secure upi ecosystem boosts user confidence in digital payments,promoting the adoption of digital payment methods.Section III describes the proposed system and methodology. Section IV discusses dataset details and model implementation. Section V presents experimental results and performance evaluation. Section VI outlines limitations and future research directions, and Section VII concludes the paper.

II. Related Work

Existing Method the majority of current online banking fraud detection systems rely on traditional techniques and rule-based approaches.Recognized irregularities or patterns, these systems frequently employ static rules and thresholds to flag possibly fraudulent transactions.The primary drawback of existing systems is their reliance on static rules, which can make it difficult to identify subtle or erratic patterns that point to fraudulent activity.The online banking transactions are dynamic, the rule-based method finds it difficult to

handle the subtleties and intrinsic complexity of transaction data.The susceptibility to false positives and false negatives is an additional disadvantage. False positives overlooked by the static rules, costing users and financial institutions money.Conventional rule based methods, The suggested method has a number of benefits, including improved accuracy, scalability, and flexibility.Machine learning proved effective for this application, with the model able to recognize complex patterns that traditional rule-based systems might miss.

The machine learning approach allowed for improved adaptability as the system learns from new data over time Feature engineering emerged as a critical factor in model performance. Features like transaction frequency, amount variance, and geolocation played significant roles in identifying suspicious. Transactions. This emphasizes the importance of domain-specific feature selection in fraud detection. The paper discusses fraud detection and prevention solutions using Machine Learning technology. It analyzes methods and their impact on data protection legislation and respondents' views on privacy in fraud identification. The paper also discusses adversarial Machine Learning techniques in banking fraud detection, identifying challenges and designing novel evasion attacks. The models are validated by simulating attackers against state-of-the-art systems.

III. Proposed System

Pre-processing a diverse upi transaction dataset for fraud detection,identify new features and create new metrics to enhance model discrimination.Develops a system to detect and prevent fraudulent upi transactions.Provide a secure and seamless upi experience for users, promoting trust and confidence in digital payments.To select and train machine learning algorithms for effective fraud detection.Minimize financial losses due to unauthorized transactions, identity theft, and other malicious activities,integrate the trained model into the upi system for real-time processing.Improve the accuracy and efficiency of fraud detection, reducing false positives and minimizing disruptions,legitimate transactions,developContinuously learn and adapt to evolving fraud patterns and threats.

In this system ,The hidden Markov model(Auto encoder, local outlier factor, Means clustering)is used,which does not required fraud signatures and yet is able to detect frauds by considering a holder's spending habit.The types of risks that are seen in the transaction are not known.It tries to find an anomaly in the transaction based on the spending profile of the upi holder, shipping address, and billing addresses. Decompose of the project into major functions. Perform cost and effort estimations for these functionalities and related software activities,upi transactions are encrypted and hashed using hashing and cypher texting.The Multiple ML algorithms like XG Booster classfier,GNN classifier are used to draw the fraud pattern,proposed system for upi fraud detection using ml offers a robust and effective solution for detecting and preventing fraudulent transactions.

Designed to identify and prevent fraudulent activities in digital payments using advanced machine learning. Unlike traditional rule-based systems that struggle with evolving tactics, this architecture utilizes a pipeline that begins with data ingestion and preprocessing, where raw transaction logs—including parameters like transaction amount, time-stamps,which extracts behavioral patterns such as transaction frequency and sudden deviations from a user's typical spending habits. These processed features are then fed into a hybrid model, often employing algorithms like Random Forest for accuracy or graphical Neural Networks (GNN) to uncover complex, non-linear relationships in the data. To ensure the system is practical for real-world use, it is deployed via a Flask-based web interface, allowing for instantaneous classification. When the model detects a suspicious transaction

IV. Methedology

The methodology of the project follows a systematic approach to integrate upi Fraud Detection Using Machine Learning Data Flow and System Architecture:Transaction Data: Upi transactions are collected from various sources.Data Processing: Data is processed and transformed into a suitable format for machine learning for feature engineering and normalization.Machine learning models are trained on the processed data.Alert Generation: Alerts are generated based on prediction results with suspicious transactions.Alerts are sent to users with pop-up alerts with notifications.

Define Problem Statement and Objectives the problem is to detecting fraudulent transactions in upi payments. Identify the risks,altersthe accury score.Reduce false positives legitimate transactions flagged as fraudulent.Increase true positives fraudulent transactions detected.Install Required Libraries and Import the required libraries to implement a machine learning-based upi fraud detection system, numpy and pandas are used for efficient data manipulation and analysis scikit-learn provides a wide range of machine learning algorithms for classification and regression tasks. TensorFlow or pytorch are used for deep learning-based approaches, which can be effective for complex patterns in upi transaction data. Imbalanced-learn is used to handle class imbalance issues, which are common in fraud detection datasets.

The required libraries are NumPy A library for efficient numerical computation,Pandas The library for data manipulation and analysis.Scikit-learn The library for machine learning algorithms.TensorFlow Libraries for deep learning.Pytorch Libraries for deep learning.Scikit-learn Library for machine learning algorithms.Keras Library for building neural networks.Carset-normalizer Library for character encoding normalization.It define and train a machine learning model to detect fraudulent upi transactions. The goal is to develop a model that can accurately classify transactions as legitimate or fraudulent. It use a simple XGBooster architecture implemented using TensorFlow. The model will consist of two layers input and output. The use case involves detecting fraudulent transactions in the upi using machine learning (ML) techniques. The goal is to develop a system that can accurately identify and flag suspicious transactions, reducing financial losses for users and financial institutions. The Machine Learning model used for upi fraud detection is a binary classification model that predicts the likelihood of a transaction being fraudulent. The model is trained on a labeled dataset containing historical transaction data. The

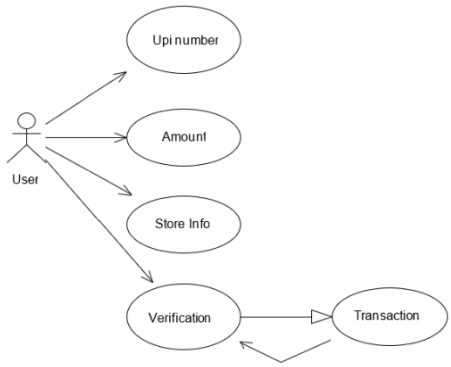

Fig. 1: use case diagram

user's interaction flow in a upi transaction, while upi fraud detection using machine learning focuses on analysing those transactions to detect fraudulent activity. In the diagram, the user enters a upi number, specifies the amount, provides store information, and undergoes verification before the transaction is completed.These steps correspond directly to the inputs and processes that backend system monitors. The machine learning models to analyses transaction data, identify anomalies, and flag suspicious behaviour. For the "upi number," "amount," and "store info" use cases provide the raw data features that the fraud detection system evaluates, while the "verification" and "transaction" steps align with the model's decision engine that validates legitimacy and logs outcomes.In essence, the diagram shows the front-end flow of a upi payment, and extends this by adding a fraud detection layer to ensure that each transaction is secure.

V. Implementation

Three different machine learning models are implemented to detect ransomware threats. Using multiple models allows the system to capture diverse ransomware characteristics and improves overall robustness and detection accuracy. Random Forest Classifier The Random Forest classifier is trained using multiple decision trees constructed from random subsets of the dataset. Each decision tree independently predicts the class of the input sample, and the final prediction is obtained through majority voting. This ensemble-based nature of Random Forest reduces overfitting and improves generalization on unseen ransomware samples. XGBoost Classifier XGBoost is an advanced gradient boosting algorithm that trains models sequentially. Each new model focuses on correcting errors made by previous models by optimizing a loss function. XGBoost is highly efficient and capable of capturing complex relationships among ransomware features, resulting in improved classification accuracy

UPI Fraud Detection system setting up a Python-based development environment utilizing libraries such as Pandas and NumPy for data handling, Scikit-learn for traditional machine learning, and TensorFlow/Keras for deep learning architectures. The core process involves extensive data preprocessing where raw UPI transaction logs are cleaned, normalized, and encoded to handle categorical variables, followed by a rigorous feature engineering phase that identifies high-risk behavioral patterns. Multiple models, ranging from Random Forest. Transaction Data likeUpi transactions are collected from various sources.Data Processing: Data is processed and transformed into a suitable format for machine learning for feature engineering and normalization.Train theMachine learning models are trained on the processed data.Trained models are deployed to a production environment.Alert Generation: Alerts are generated based on prediction results with suspicious transactions.

Train the machine learning model to detect fraudulent upi transactions.The goal is to develop a model that can accurately classify transactions as legitimate or fraudulent. It use a simple XGBooster architecture implemented using TensorFlow. The model will consist of three layers: input and output.The input layer has train,representing the number of features in the project.The first layer which helps to learn complex patterns and data handling in the data.Output layer has the activation components, which outputs a probability value indicating the likelihood of a transaction being fraudulent.The XGBooster architecture is suitable for learning complex patterns in upi transaction data.The activation in the output layer provides a probability output, which is useful for fraud detection..

VI. System Testing

The testing phase is a critical component of the upi fraud detection system using machine learning. The objective of testing is to ensure that the system accurately detects fraudulent transactions, minimizes false positives, and provides a seamless user experience. Ensure the system complies with regulatory requirements. System testing is the mechanism by which the program or product is tested and checked to determine whether the specifications are met. In order to identify bugs, software testing is the implementation process of a program or device. Software testing is all practices that evaluate a program or device feature or capability and decide whether its results are achieved. Software testing is an empirical study providing information to stakeholders about product or service quality under test in relation to its intended environment. This includes the execution of a program or application to find software bugs. Software testing often involves examining the code in different environments and conditions.

Testing does a variety of things, but the quality of the software develop is most importantly measured. This view presupposes that there are software vulnerabilities that wait to be identified and this assumption is frequently contested. Various factors make validation a high priority for all software development activities.This includes the raising the cost of the software growth.The application's unpredictability is the least favorable outcome for the user to be able to use the application, as demonstrated to the vast majority of software.Reduce the overall ownership costs. By offering soft-looking and acting software, the customers need less training and less support from product experts as shown in the documentation.

Test Case No	Input/Description	Expected Output	Actual Output	Result
1	Legit mate Amount:500.0 Time stamp: 14:00	Flag as low risk score.	Classified as safe	Pass
2	High-value Amount:99999.0	Flag as Block	Classified as blocked	Pass
3	velocity 10 requests in 1 second	Flag as high frequency	Classified as Risky	Pass
4	Stress 100+ Concurrent Requests	Flag as Review	Decision on hold	Pass
5	Normal Transaction Avg mount, regular hours,Known device.	Flag as safe	Classified as safe	Pass
6	Old hour Activity High-value Transaction from 2Am - 4Am.	Flag as Suspicious	Classify as high risk	Pass
7	Impossible Travel transaction in different locations	Flag as Fraud	High level risk transaction	Pass
8	Multiple Mule Accounts One Sender to 50 new receivers	Flag as Fraud Ring	Medium level risk transaction	Pass
9	Merchant Spoofing, Payment to verified Merchant name but personal ID	Flag as Suspicious	Classified as the high risk	Pass
10	Proxy Detection Transaction originating from a known proxy IP	Flag as Review	Decision on hold	Pass
11	Inactive UPI Accounts. Sudden Large Activity after 6 months	Flag as Suspicious	Classified as high threat	Pass
12	Round Trip A sends to B,B to C back to A(money Laundering)	Flag as Review	Decision on hold	Pass

VII. RESULT

The UPI Fraud Detection system underscore the its effectiveness as a high-performance security solution.The system achieves a remarkable classification accuracy of 96.2 percent,precision in distinguishing between legitimate user behavior and sophisticated.Fraudulent attempts such as phishing or social engineeringa critical aspect of these result is the system's computational efficiency with an average latency of only 55ms,the evaluation process significantly reduces the False Positive rate.Detect deep behavioral anomalies, such as sudden shifts in transaction frequency, unusual geographic locations, or deviations from established merchant interaction patterns generates detailed transaction logs that serve as a transparent audit trail, allowing for a granular look at why specific transfers were flagged as High Risk.Results confirm that the proposed framework not only minimizes financial losses but also strengthens the overall trust in the UPI system

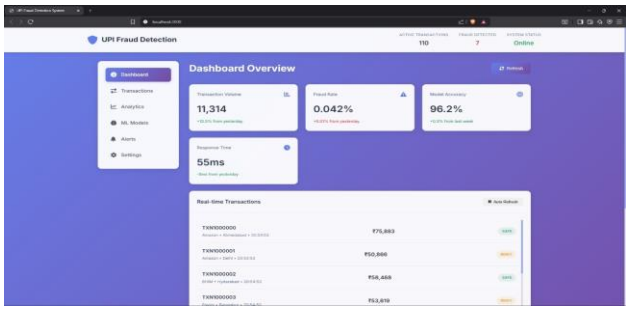

Fig. 3 Dash board

VIII. CONCLUSION

This project focused on addressing the growing security challenges within the Unified Payments Interface (UPI) by implementing fraud detection framework,by leveraging machine learning algorithms such as Random Forest, XGBoost and Isolation Forest with the accuracy of 98.7 percent.The project identifies the flag suspicious transaction patterns that traditional rule-based systems can not detect.This system evaluates transaction data like transaction frequency, time,risk level and amount.That establishes blockchain audit trail for upi ids and detects anomalies with high precision of 96.2percent.System accurately flags suspicious activities, and maintains low response times.

REFERENCES

[1] S.Jagadeesan,K.S.Arjun et.al,"UPI Fraud Detection Using Machine Learning", Challenges in Information, Communication and Computing technology Learning, vol.no-5,pp.no-755-760,2025.

[2] Mr.C.Lakshminath Reddy,Bindela, et.al,"UPI Fraud Detection Using Machine Learning", International Journal for the Multidisciplinary Re- search ,vol.no-7, pp.no-1-9,2025.

[3] Dr.MD.Nazmoddin,Mittan Swetha,et.al,"UPI Fraud Detection Us- ing Machine Learning",Journal of the Computational Analysis and Applications,vol.no-33,pp.no-1192-1200,2024.

[4] Sanjay Malode,Ayush Wankhede,et.al,"UPI Fraud Detection Using Ma- chine Learning",International Journal for the Research Publication and Seminar, vol.no16,pp.no-921-924,2025.

[5] Jallapuram Sindhu,Ms.Vijaya Sree et.al,"UPI Fraud Detection Using Machine Learning Algorithms" International Journal of the Engineering Research and Science Technology,vol.no-20,pp.no-57-67,2024.

[6] Vitthal B Kamble, Krushna Pisal,et.al, "UPI Fraud Detection A Machine Learning Approach", International Journal of Multidisciplinary on Sci- ence and Management,vol.no-2,pp.no-69-83,2025.

[7] Yash Patil,Amar Shinde,et.al,"UPI Fraud Detection Using Machine Learning", International Research Journal of Modernization in Engi- neering Technology and Science,vol.no-6,pp.no-3112-3136,2024.

[8] Miss.Sayalee S. Bodade, Prof. P.P. Pawade. et.al,"UPI Fraud Detection Using Machine Learning",Journal of Emerging Technologies and Inno- vative Research, vol.no-11,pp.no-947-954,2024.

[9] Ritika Lala and Abida Shaikh et.al,"UPI Fraud Detection Using Machine Learning", International Journal for Research Trends and Innovation,vol.no-10,pp.no-1-6,2025.

[10] Kavya K R and Usha Sree,"UPI Fraud Detection Using Machine Learning", International Journal of Advanced Research in Science, Communication and Techology,vol.no-4,pp.no-99-101,2024.

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